

Rangkuman materi: PK dan PM

I. Fungsi Kuadrat

Bentuk umum

$$y = f(x) = ax^2 + bx + c, a \neq 0$$

Titik puncak / ekstrem / min / maks

$$x_p = \text{sumbu simetri} = \frac{-b}{2a} \text{ (absis)}$$

$$y_p = \text{nilai ekstrem} = \frac{D}{-4a} \text{ (ordinat)}$$

Menyusun persamaan kuadrat

Jika diketahui:

1. tiga titik Jembarang

$$y = ax^2 + bx + c \text{ (eliminasi)}$$

2. titik puncak

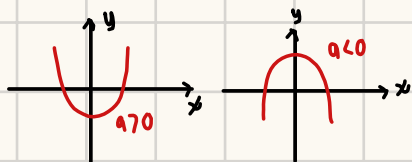
$$y - y_p = a(x - x_p)^2$$

3. titik potong sb x

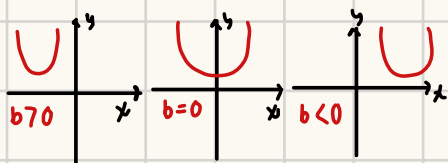
$$y = a(x - x_1)(x - x_2)$$

Hubungan a, b, c, dan D dengan grafik

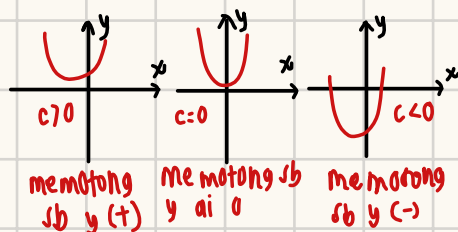
• a



• b



• c

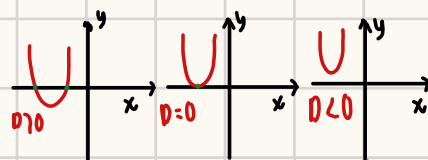


memotong sb y (+)

memotong sb y di 0

memotong sb y (-)

• D



memotong sb x di dua titik.

menyentuh sb x

tdk memotong sb x

Definit

Definit (+) : $a > 0, D < 0$

Definit (-) : $a < 0, D < 0$

II. Persamaan Kuadrat

Akar-akar:

$$x_{1,2} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Operasi akar-akar

- $x_1 + x_2 = -b/a$
- $x_1 x_2 = c/a$
- $x_1 - x_2 = \pm \sqrt{D}/a$
- $x_1^2 + x_2^2 = (x_1 + x_2)^2 - 2x_1 x_2$
- $x_1^3 + x_2^3 = (x_1 + x_2)^3 - 3x_1 x_2 (x_1 + x_2)$
- $1/x_1 + 1/x_2 = \frac{x_1 + x_2}{x_1 x_2}$
- $x_1^2 - x_2^2 = (x_1 + x_2)(x_1 - x_2)$

Sifat akar-akar

- Dua akar positif: $x_1 + x_2 > 0; x_1 x_2 > 0; D > 0$
- Dua akar negatif: $x_1 + x_2 < 0; x_1 x_2 > 0; D > 0$
- Salang berlawanan: $x_1 x_2 < 0; D > 0$
- Salang berkebalikan: $x_1 x_2 = 1; D > 0$

III. Fungsi Komposisi dan Invers

Daerah asal / domain.

$$a) y = \frac{f(x)}{g(x)} \rightarrow g(x) \neq 0$$

$$b) y = \sqrt{f(x)} \rightarrow f(x) \geq 0$$

Daerah hasil / range

$$a) y = ax + b \rightarrow -\infty < y < \infty$$

$$b) y = \sqrt{f(x)} \rightarrow f(x) \geq 0$$

Sifat-sifat

$$a) f \circ g(x) = f(g(x))$$

$$b) f \circ g(x) = h(x) \begin{cases} f(x) = (h \circ g^{-1})(x) \\ g(x) = (f^{-1} \circ h)(x) \end{cases}$$

$$c) (f^{-1}(x))^{-1} = f(x)$$

$$d) f^{-1} \circ f(x) = f \circ f^{-1}(x) = f(x)$$

$$e) f^{-1}(a) = b \rightarrow f(b) = a$$

$$f) (f \circ g)^{-1}(x) = g^{-1} \circ f^{-1}(x)$$

Invers dari fungsi

$$a) f(x) = ax + b$$

$$f^{-1}(x) = \frac{x-b}{a}$$

$$b) f(x) = \frac{ax+b}{cx+a}$$

$$f^{-1}(x) = \frac{-dx+b}{cx-a}$$

$$c) f(x) = ax^2 + bx + c$$

$$f^{-1}(x) = \frac{-b \pm \sqrt{b^2 - 4a(c-x)}}{2a}$$

IV. Statistika

Mean / rata-rata ($\bar{x}$)

↳ jumlah semua data dibagi banyaknya data.

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n}$$

Rata-rata gabungan

$$\bar{x}_{gab} = \frac{\bar{x}_1 \cdot n_1 + \bar{x}_2 \cdot n_2}{n_1 + n_2}$$

Median / Nilai Tengah (me)

• Data ganjil • Data genap

$$\hookrightarrow me = \frac{x_{n+1}}{2} \quad \hookrightarrow me = \frac{x_{\frac{n}{2}} + x_{\frac{n}{2}+1}}{2}$$

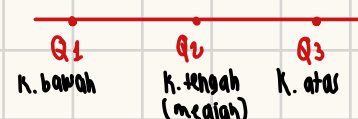
Modus (Mo)

Nilai yang paling sering muncul.

Jangkauan

Data terbesar - data terkecil

Kuartil



V. Baris dan deret.

Baris Aritmatika

↳ Barisan yang tiap bilangannya mempunyai selisih yang tetap.

- suku ke-n

$$u_n = a + (n-1)b$$

$$b = u_2 - u_1$$

- suku tengah

tiga suku berurutan membentuk aritmatika,

maka berlaku:

$$2u_2 = u_1 + u_3$$

Deret Aritmatika

↳ Jumlah dari suku-suku aritmatika

$$S_n = \frac{n}{2}(2a + (n-1)b)$$

$$S_n = \frac{n}{2}(a + u_n)$$

$$S_n = n \cdot u_t \text{ (suku tengah)}$$

- Jumlah n suku pertama berbentuk

$$S_n = pn^2 + qn$$

$$a = p + q$$

$$b = 2p$$

$$u_n = S_n - p$$

Baris geometri

↳ barisan bilangan yang memiliki aturan perbandingan dua suku tetap.

- suku ke-n

$$u_n = ar^{n-1}$$

$$r = \frac{u_1}{u_2}$$

- suku tengah

$$u_2^2 = u_1 \times u_3$$

Deret geometri

- $r < 1 \rightarrow S_n = a \frac{1-r^n}{1-r}$

- $r > 1 \rightarrow S_n = a \frac{r^n - 1}{r - 1}$

Deret geometri tak hingga

$$S_\infty = \frac{a}{1-r}$$

Jumlah tak hingga suku genap/ganjil

- genap = $\frac{ar}{1-r^2}$

- ganjil = $\frac{a}{1-r^2}$

Panjang lintasan bola yang dipantulkan

$$S = \left| \frac{a+p}{q-p} \right| h_0$$

$$\text{rasio} = p/q$$

VI. Persamaan garis lurus

Bentuk umum

$$ax + by = c$$

$$y = mx + c$$

Menentukan persamaan garis lurus

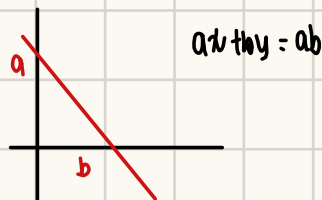
a) Diketahui satu titik (x_1, y_1)

$$y - y_1 = m(x - x_1)$$

b) Diketahui dua titik (x_1, y_1) dan (x_2, y_2)

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

c) memotong kedua sumbu



Menentukan gradien

- $y = mx + c$, gradien = m

- $ax + by = c$, gradien = $-\frac{a}{b}$

- Diketahui dua titik

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

- Diketahui sudut

$$m = \tan \theta$$

Hubungan antara dua garis

- sejajar (tidak punya solusi) $\neq$

$$m_1 = m_2$$

- tegak lurus $\perp$

$$m_1 \cdot m_2 = -1$$

- berimpit (banyak solusi)

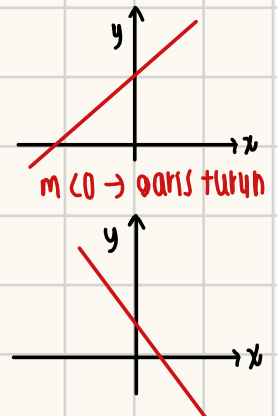
$$m_1 = m_2 \text{ dan } c_1 = c_2$$

- berpotongan

$$m_1 \neq m_2$$

Tanda gradien

$m > 0 \rightarrow$ garis naik



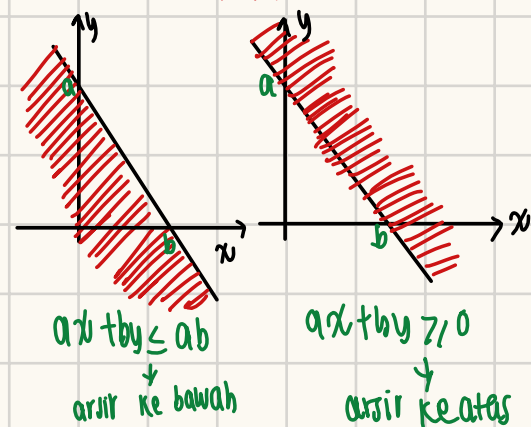
Jarak titik ke garis

Diketahui persamaan garis $ax + by$

+ c

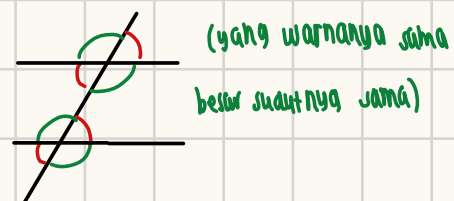
$$d = \left| \frac{ax_0 + by_0 + c}{\sqrt{a^2 + b^2}} \right|$$

VII. Pertidaksamaan linear



VIII. Geometri sudut dan bidang datar.

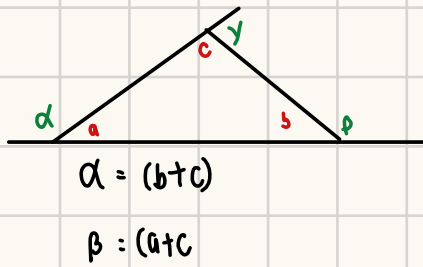
Hubungan dua garis sejajar



Segitiga

- Jumlah sudut = 180°

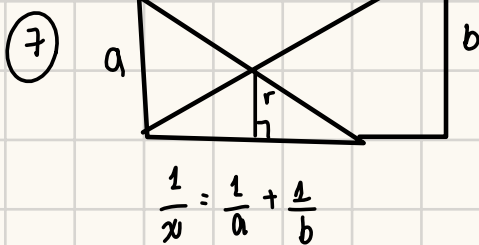
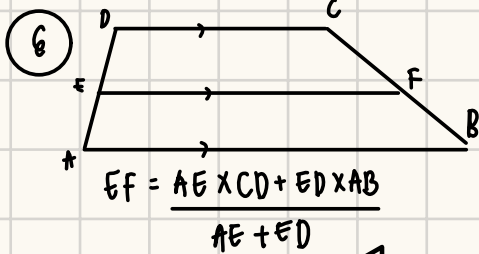
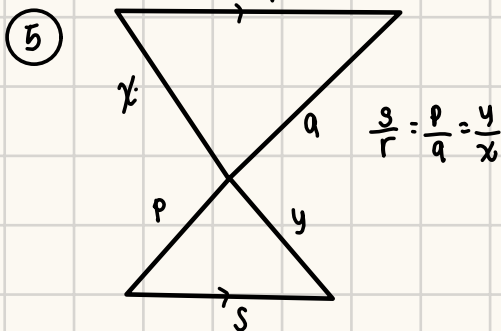
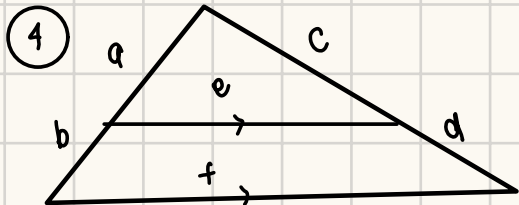
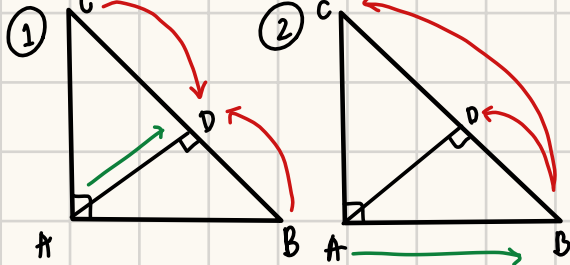
• sudut luar segitiga



• Luas segitiga

$$L = \frac{1}{2} \times \text{alas} \times \text{tinggi}$$

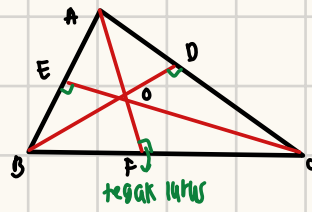
• kesebangunan pada segitiga



• garis istimewa pada segitiga

a) garis tinggi

↳ ditarik dari sudut segitiga menuju sisi dihadapannya.

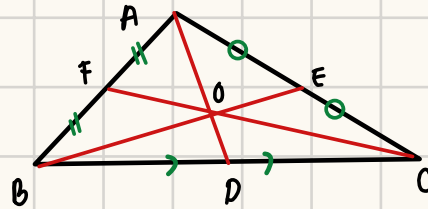


$$AF : CE : BD = \frac{1}{BC} : \frac{1}{AB} : \frac{1}{AC}$$

sisi yg ada dihadapannya.

b) garis berat segitiga

↳ garis yg ditarik dari sudut segitiga ke sisi dihadapannya dan membagi dua sisi sama besar.

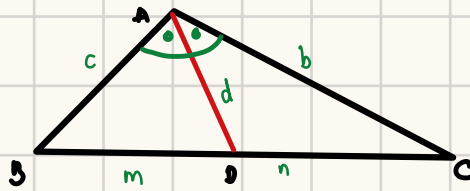


$$\frac{AO}{OD} = \frac{CO}{OF} = \frac{BO}{OE} = \frac{2}{1}$$

$$AD^2 = \frac{1}{2} AB^2 + \frac{1}{2} AC^2 - \frac{1}{2} BC^2$$

c) garis bagi segitiga

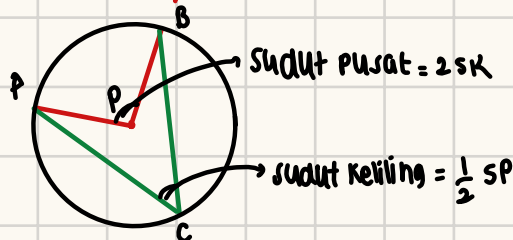
↳ garis yang ditarik dari titik sudut segitiga ke sisi dihadapannya menjadi dua sudut sama besar.



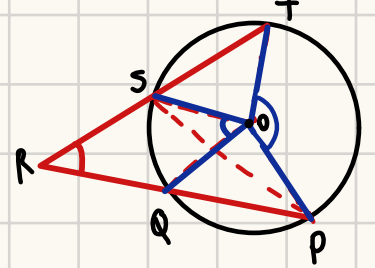
$$d^2 = (c \times b) - (m \times n)$$

Lingkaran

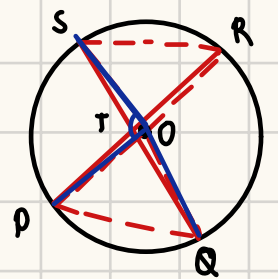
• sudut pusat dan sudut keliling



• sudut dua tali busur dalam lingkaran



$$\angle TRP = \frac{1}{2} (\angle TOP - \angle SOQ)$$



$$\angle PTS = \frac{1}{2} (\angle POS + \angle QOS)$$

• Luas keliling lingkaran

$$K = \pi d = 2\pi r$$

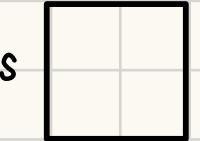
$$L = \frac{1}{4} \pi d^2 = \pi r^2$$

Segiempat

• persegi

$$L = s^2$$

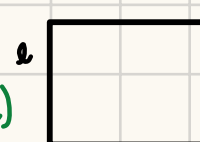
$$K = 4s$$



• persegi panjang

$$L = p \times l$$

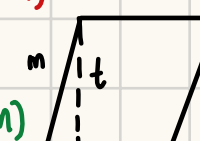
$$K = 2(p+l)$$



• jajargenjang

$$L = a \times t$$

$$K = 2(a+t)$$



• belah ketupat

$$L = \frac{1}{2} d_1 \times d_2$$

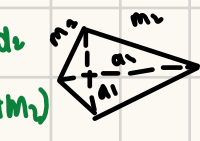
$$K = 4m$$



• Layang-layang

$$L = \frac{1}{2} d_1 \times d_2$$

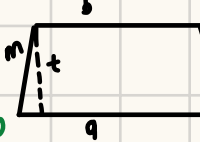
$$K = 2(m_1 + m_2)$$



• Trapezium

$$L = \frac{1}{2} (a+b) \times t$$

$$K = 2m + a + b$$

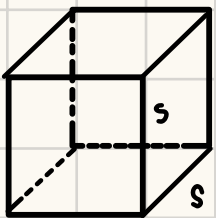


Simetri lipat dan simetri putar

Bangun datar	s. lipat	s. putar
persegi	4	4
p. panjang	2	2
Δ sama sisi	3	3
Δ sama kaki	1	1
Δ siku ^x	0	1
jajargenjang	0	2
belah ketupat	2	2
Trapezium siku ^x	0	1
Trapezium sm kaki	1	1
Layang-layang	1	1
Lingkaran	∞	∞

IX. Bangun Ruang

a) kubus



ciri: 12 rusuk sama panjang
6 sisi berbentuk persegi

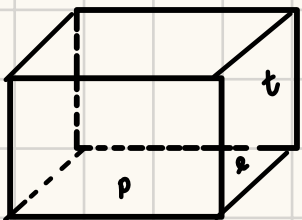
$$s \text{ Luas permukaan} = 6 \times s^2$$

$$\text{Volume} = s^3$$

$$\text{Diagonal sisi} = s\sqrt{2}$$

$$\text{Diagonal ruang} = s\sqrt{3}$$

b) balok



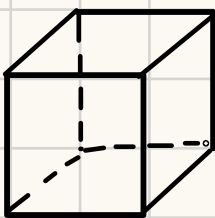
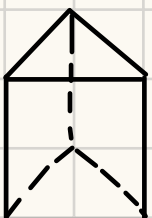
ciri: sisi → persegi panjang
3 pasang persegi panjang

$$\text{Luas permukaan} = 2(pl + pt + lt)$$

$$\text{Volume} = p \times l \times t$$

$$\text{Diagonal ruang} = \sqrt{p^2 + l^2 + t^2}$$

c) prisma

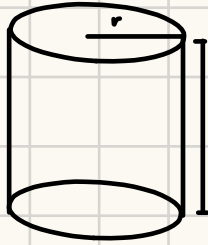


ciri: alas dan atap merupakan bid. datar yg sama

$$\text{Luas permukaan} = 2 \times \text{L. alas} + \text{luas selimut}$$

$$\text{Volume} = \text{L. alas} \times t$$

d) Tabung/silinder



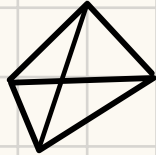
ciri: alas berupa lingkaran

$$\text{L. selimut} = 2\pi r t$$

$$\text{L. permukaan} = 2\pi r(r+t)$$

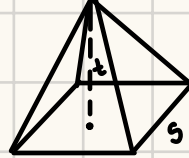
$$\text{Volume} = \pi r^2 t$$

e) limas



limas Jajargenjang

limas Jajargenjang



$$\text{L. permukaan} = \text{L. alas} + \text{jumlah L. sisi tegak}$$

$$V = \frac{1}{3} \text{LA} \times t$$

f) Kerucut

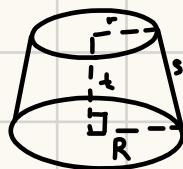


ciri: garis pelukis

$$\text{L. selimut} = \pi r s$$

$$\text{Volume} = \frac{1}{3} \pi r^2 t$$

Kerucut terpancung



$$\text{Volume} = \frac{1}{3} \pi t (R^2 + Rr + r^2)$$

$$\text{L. permukaan} = \text{L. alas} +$$

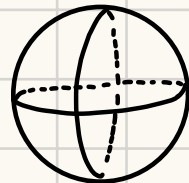
$$\text{L. tutup} + \text{L. selimut}$$

$$\text{L. alas} = \pi R^2$$

$$\text{L. tutup} = \pi r^2$$

$$\text{L. selimut} = \pi s (R+r)$$

g) Bola



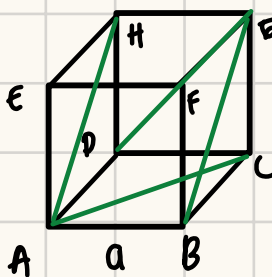
$$\text{L. permukaan} = 4\pi r^2$$

$$\text{Volume} = \frac{4}{3} \pi r^3$$

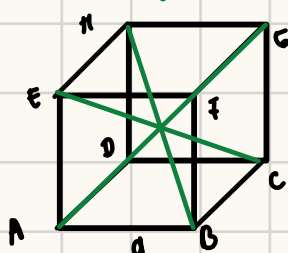
X. Dimensi Tiga

a) kubus

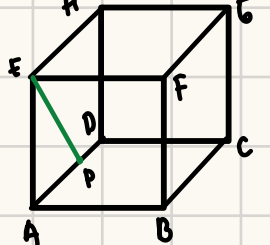
- diagonal bidang $a\sqrt{2}$



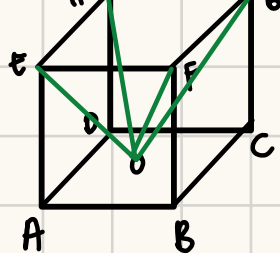
- diagonal ruang $a\sqrt{3}$



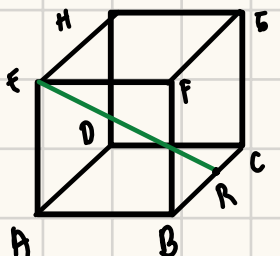
- jarak EP $\frac{1}{2} a\sqrt{5}$



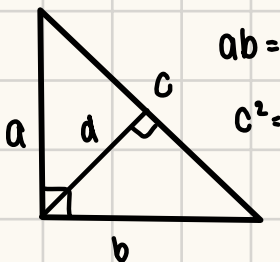
- jarak GO, HO, EO, FO $\frac{1}{2} a\sqrt{6}$



- jarak ER



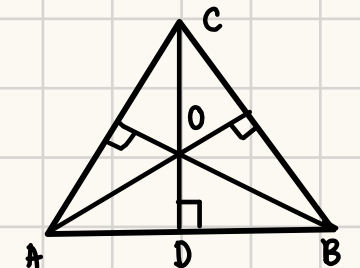
b) segitiga siku-siku



$$ab = cd$$

$$c^2 = a^2 + b^2$$

c) segitiga sama sisi



$$CD = \frac{1}{2} s\sqrt{3}$$

$$CO = \frac{1}{3} s\sqrt{3}$$

$$OD = \frac{1}{6} s\sqrt{3}$$

XI. Pangkat, akar, dan logaritma

a) sifat eksponen

$$- a^m \cdot a^n = a^{m+n}$$

$$- \frac{a^m}{a^n} = a^{m-n}$$

$$- (a^m)^n = a^{mn}$$

$$- a^n \cdot b^n = (ab)^n$$

$$- a^{-n} = \frac{1}{a^n}$$

$$- a^0 = 1$$

b) persamaan eksponen

- $a^{f(x)} = b^{f(x)} \rightarrow f(x) = 0$
- $a^{f(x)} = a^{g(x)} \rightarrow f(x) = g(x)$
- $a^{f(x)} = b^{g(x)} \rightarrow f(x) = g(x) \cdot \log a$
- $f(x)^{g(x)} = f(x)^{h(x)}$ memenuhi:
 1. $g(x) = h(x)$
 2. $f(x) = 1$
 3. $f(x) = -1$; $g(x)$ dan $h(x)$ keduanya sama^x ganjil/ genap
 4. $f(x) = 0$; $g(x)$ dan $h(x) \neq 0$

c) sifat-sifat akar

- $\sqrt[n]{a} = a^{1/n}$
- $\sqrt[n]{a} = a^{1/n}$
- $a\sqrt[n]{c} + b\sqrt[n]{c} = (a+b)\sqrt[n]{c}$
- $\sqrt[n]{a} \cdot \sqrt[n]{a} = a$
- $\sqrt[n]{a} \cdot \sqrt[n]{b} = \sqrt[n]{ab}$
- $\sqrt[n]{a^m} \cdot \sqrt[n]{a^p} = \sqrt[n]{a^{m+p}}$
- $\sqrt[n]{\sqrt[n]{a}} = \sqrt[n]{a}$

d) akar tak hingga

- $\sqrt{a\sqrt{a\sqrt{\dots}}} = a$ ^{ambil x nya aja}
- $\sqrt{a:\sqrt{a:\sqrt{\dots}}} = x^3$ ^{ambil x nya aja}
- $\sqrt{a+\sqrt{a+\sqrt{\dots}}} =$ faktor selisih 1 yang besar
- $\sqrt{a-\sqrt{a-\sqrt{\dots}}} =$ faktor selisih 1 yang kecil
- $\sqrt{ax+\sqrt{ax+\sqrt{\dots}}} = b$ ^{katikan hasil dgn angka jtlhnya (misal c)}
 $\hookrightarrow ax = bc$
 $x =$
- $\sqrt{1+n\sqrt{1+(1+n)\sqrt{\dots}}} = a$
 $1+n=a \rightarrow n = \dots$

e) Logaritma

- $\hookrightarrow$ invers dari eksponen
- $a^n = b \Rightarrow \log_a b = n$
^{basis}

f) sifat-sifat logaritma

- $\log_a b + \log_a c = \log_a (bc)$
- $\log_a b - \log_a c = \log_a \left(\frac{b}{c}\right)$
- $\log_a b^n = n \cdot \log_a b$
- $\log_a b = \frac{1}{\log_b a}$
- $\log_a b = \frac{\log b}{\log a}$
- $\log_a a = 1$
- $\log_a b \cdot \log_b c = \log_a c$
- $a^{\log_a c} = c$

g) persamaan logaritma

- $\log_a f(x) = p \rightarrow f(x) = a^p$
- $\log_a f(x) = \log_a g(x) \rightarrow f(x) = g(x)$
- $\log_a f(x) = \log_b g(x) \rightarrow f(x) = g(x) = 1$
- $a(p \log x)^2 + b(p \log x) + c = 0$
 $\hookrightarrow x_1 \cdot x_2 = p^{-b/a}$

XII. Bilangan dan operasinya

a) jenis bilangan

- bil. genap $\rightarrow$ habis dibagi dua
- bil. ganjil $\rightarrow$ tidak habis dibagi dua
- bil. prima $\rightarrow$ bilangan yang hanya dapat dibagi 1 atau angka itu sendiri.
cth: 2, 3, 5, 7, 11, 13, dst.
- bil. komposit $\rightarrow$ bil. lebih dari satu dan bukan bil. prima
cth: 4, 6, 8, 9, 10, 12, dst.
- bil. kuadrat: bil. hasil perkalian dua bil. yang sama.
cth: 4, 9, 16, dst.

b) pembagian bilangan.

- **habis dibagi 3**: jumlah semua digit dalam bilangan habis dibagi 3. cth: 2418 habis dibagi 3? ya
 $\hookrightarrow 2+4+1+8 = 15 \rightarrow$ habis dibagi 3
- **habis dibagi 4**: dua digit terakhir bilangan habis dibagi 4. cth: 56324
 $\hookrightarrow 24 \rightarrow$ habis dibagi 4.

- **habis dibagi 5**: digit terakhir bilangannya 0 / 5

- **habis dibagi 6**: genap & jumlah semua digit habis dibagi 3.

cth: 2418 $\rightarrow$ genap

$\hookrightarrow 2+4+1+8 = 15$

habis dibagi 3

- **habis dibagi 7**:

kalikan digit satuan

dgn 2, lalu hasilnya meng-

urangi pengurang sisa digit.

Jika hasil pengurangan

habis dibagi 7, bil. tsb

habis dibagi 7.

cth: 43778, digit satuan

$43778 - 2(8) = 4361$

^{selain digit satuan}

$436 - 2(1) = 434$

$43 - 2(4) = 35$

$\downarrow$
habis dibagi 7

- **habis dibagi 8**: tiga

digit terakhir habis dibagi 8.

- **habis dibagi 9**: jumlah digit habis dibagi 9

- **habis dibagi 11**:

jml digit pd posisi ganjil

dikurangi jml digit pd

posisi genap = 0.

cth: 9372

$9+7 = 16$

$3+2 = 5$
 $\underline{\quad 11 \quad}$

- habis dibagi 13 : kalikan digit duaannya XIV. Jarak, kecepatan, dan waktu.

dengan 9, hasilnya menjadi pengurang
sisa digitnya. jika hasil pengurangan habis
dibagi 13 maka bil. tsb habis dibagi 13.

$$cth: 3419$$

$$341 - 9(9) = 260$$

$$26 - 9(0) = 26$$

↳ habis dibagi 13.

c) Kelipatan persekutuan kecil (KPK)

cth: KPK dari 4 dan 5

Kel. 4 = 4, 8, 12, 16, 20, 24, 28, ...

Kel. 5 = 5, 10, 15, 20, 25, ...

KPK = 20 → kelipatan yg sama dan terkecil

a) faktor persekutuan besar (FPB)

cth: FPB dari 20 dan 12

Faktor 20: 1, 2, 4, 5, 10, 20.

Faktor 12: 1, 2, 3, 4, 6, 12.

FPB: 4 → faktor sama yg paling besar.

XIII. Aritmatika sosial

a) Keuntungan (%)

$$\frac{\text{Harga jual} - \text{Harga beli}}{\text{Harga beli}} \times 100\%$$

b) Kerugian (%)

$$\frac{\text{Harga beli} - \text{Harga jual}}{\text{Harga beli}} \times 100\%$$

c) Diskon

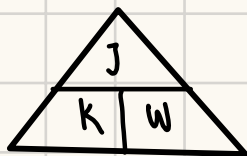
$$\text{Harga Bayar} = \text{Harga awal} \times (100\% - \% \text{ diskon})$$

a) Bunga tunggal

$$M_0 \times \% \text{ bunga} \times n$$

e) Bunga majemuk

$$M_n = M_0 (1 + \%b)^n$$



a) waktu berpapasan (jam berangkat sama)

$$t_p = \frac{S}{v_A + v_B}$$

b) waktu berpapasan (jam berangkat beda)

$$t_p = \frac{S - (\text{selisih jarak})}{v_A + v_B}$$

c) waktu menyusul (waktu berangkat beda)

$$t_m = \frac{\text{selisih jarak}}{v_B - v_A}$$

XV. Kaidah pencacahan dan peluang

a) aturan perkalian

↳ sesuatu terjadi dengan n tahap berurutan, tahap pertama terdapat a_1 tahap berbeda sampai tahap ke-n.

$$\text{Total kejadian: } a_1 \times a_2 \times a_3 \times \dots \times a_n$$

b) permutasi

↳ pola pengambilan yg memperhatikan urutan.

cth: jabatan, cara duduk, dll.

$$\text{- unsur beda } P_r^n = \frac{n!}{(n-r)!}$$

$$\text{- unsur sama } P_{n_1, n_2, \dots}^n = \frac{n!}{n_1! n_2! \dots}$$

$$\text{- unsur melingkar (siklik)} \frac{n!}{(n-1)!}$$

c) Kombinasi

↳ pola pengambilan yg tdk memperhatikan urutan

$$C_r^n = \frac{n!}{(n-r)! r!}$$

d) Peluang

$$\text{- } P(A) = \frac{n(A)}{n(S)} \rightarrow \text{banyaknya kejadian}$$

$$\text{- } n(S) \rightarrow \text{ruang sampel}$$

$$\text{- Komplemen: } P(A^c) = 1 - P(A)$$

- Peluang gabungan

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

↳ "atau"

- Peluang kejadian saling lepas

$$P(A \cup B) = P(A) + P(B)$$

- peluang kejadian saling bebas

$$P(A \cap B) = P(A) \cdot P(B)$$

↳ "dan"

- peluang kejadian bersyarat (A dan B tdk saling bebas)

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

e) Peluang kejadian binomial

$$P(X) = C_x^n \cdot p^x \cdot q^{n-x}$$

↳ P. kejadian yg diharapkan

n = banyaknya kejadian

x = banyaknya keberhasilan

p = peluang berhasil

q = peluang gagal

f) Frekuensi harapan (Fh)

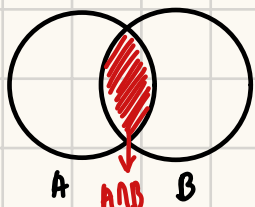
$$Fh = n \times P(A)$$

↳ byk percobaan

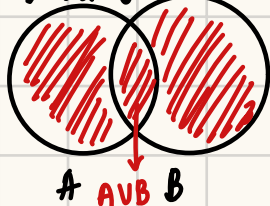
XVI. Himpunan

a) operasi himpunan

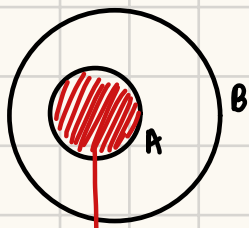
- Irisan ($\cap$)



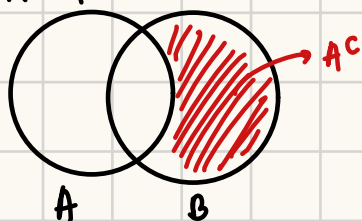
- gabungan ($\cup$)



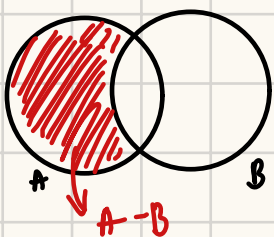
- Bagian / subset ($\subset$)



- Komplemen (A^c)



- selisih ($-$)



- $A \cup \emptyset = A$ - $A \cap (B \cap C) = (A \cap B) \cap C$
- $A \cap \emptyset = \emptyset$ - $A \cup (B \cup C) = (A \cup B) \cup C$
- $A \cup A = A$ - $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cap A = A$ - $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$
- $A \cap B = B \cap A$
- $A \cup B = B \cup A$

XVII. Matriks

a) operasi matriks

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad B = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$$

- penjumlahan / pengurangan

$$A \pm B = \begin{pmatrix} a \pm e & b \pm f \\ c \pm g & d \pm h \end{pmatrix}$$

- perkalian skalar

$$kA = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}$$

- perkalian dua matriks

$$A \cdot B = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$$

a) determinan matriks $\rightarrow$ sifat determinan

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$|A| = ad - bc$$

$$|AB| = |A| \cdot |B|$$

$$|A^T| = |A|$$

$$|A^{-1}| = \frac{1}{|A|}$$

e) Transpos matriks $\rightarrow$ sifat transpos

$$A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$A^T = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$$

$$(AB)^T = B^T A^T$$

$$(KA)^T = KA^T$$

f) Invers matriks

$$A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

sifat invers

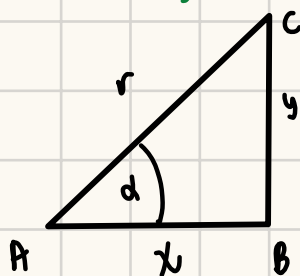
$$(AB)^{-1} = B^{-1} A^{-1}$$

$$(KA)^{-1} = \frac{1}{K} A^{-1}$$

$$(A^{-1})^{-1} = A$$

XVIII. Trigonometri

a) segitiga siku-siku



$$\sin \alpha = \frac{y}{r}$$

$$\cos \alpha = \frac{x}{r}$$

$$\tan \alpha = \frac{y}{x}$$

$$\csc \alpha = \frac{r}{y}$$

$$\sec \alpha = \frac{r}{x}$$

$$\cot \alpha = \frac{x}{y}$$

Identitas trigonometri

$$\sin \alpha = \frac{1}{\csc \alpha}$$

$$\cos \alpha = \frac{1}{\sec \alpha}$$

$$\tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{1}{\cot \alpha}$$

$$\cot \alpha = \frac{\cos \alpha}{\sin \alpha}$$

$$\sin^2 \alpha + \cos^2 \alpha = 1$$

$$\tan^2 \alpha + 1 = \sec^2 \alpha$$

$$\cot^2 \alpha + 1 = \csc^2 \alpha$$

b) sudut istimewa

$$0^\circ \quad 30^\circ \quad 45^\circ \quad 60^\circ \quad 90^\circ$$

$$\sin \alpha \quad 0 \quad \frac{1}{2} \quad \frac{1}{2}\sqrt{2} \quad \frac{1}{2}\sqrt{3} \quad 1$$

$$\cos \alpha \quad 1 \quad \frac{\sqrt{3}}{2} \quad \frac{1}{2}\sqrt{2} \quad \frac{1}{2} \quad 0$$

$$\tan \alpha \quad 0 \quad \frac{1}{\sqrt{3}} \quad 1 \quad \sqrt{3} \quad \infty$$

XIX. Transformasi geometri

a) Translasi / pergeseran

$$T \begin{pmatrix} m \\ n \end{pmatrix} \\ P(a,b) \rightarrow P'(a+m, b+n)$$

b) Refleksi / pencerminan

- thd sb x

$$(x,y) \xrightarrow{M_x} (x,-y)$$

- thd sb y

$$(x,y) \xrightarrow{M_y} (-x,y)$$

- thd garis $y=x$

$$(x,y) \xrightarrow{M_{y=x}} (y,x)$$

- thd garis $y=-x$

$$(x,y) \xrightarrow{M_{y=-x}} (-y,-x)$$

- thd titik (0,0)

$$(x,y) \xrightarrow{M_{(0,0)}} (-x,-y)$$

- thd garis $y=h$

$$(x,y) \xrightarrow{M_{y=h}} (x, 2h-y)$$

- thd garis $x=h$

$$(x,y) \xrightarrow{M_{x=h}} (2h-x, y)$$

c) Rotasi

- thd $0, 90^\circ$

$$(x,y) \rightarrow (-y, x)$$

- thd $0, -90^\circ$

$$(x,y) \rightarrow (y, -x)$$

- thd $0, 180^\circ$

$$(x,y) \rightarrow (-x, -y)$$

- thd $0, \alpha^\circ$

$$(x', y') = \begin{pmatrix} \cos \alpha & -\sin \alpha \\ \sin \alpha & \cos \alpha \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$$

d) Dilatasi

- pusat O (0,0)

$$A(a,b) \xrightarrow{(0,k)} A'(ka, kb)$$

- pusat titik P(m,n)

$$\begin{pmatrix} a' \\ b' \end{pmatrix} = k \begin{pmatrix} a-m \\ b-n \end{pmatrix} + \begin{pmatrix} m \\ n \end{pmatrix}$$